PRODUCT USER MANUAL

For Iberia-Biscay-Ireland Ocean Heat Content (0-2000m)

OMI_CLIMATE_OHC_IBI_area_averaged_anomalies

Issue: 1.1

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
OHC	Ocean Heat Content
IBI	Iberia-Biscay-Ireland
IBI-MYP	Copernicus Marine product: IBI_MULTIYEAR_PHY_005_002 (reanalysis dataset)
IBI-INT	Copernicus Marine product: IBI_MULTIYEAR_PHY_005_002 (interim dataset)
GLO-MYP	Copernicus Marine product : GLOBAL_MULTIYEAR_PHY_001_030
CORA	Copernicus Marine product: INSITU_GLO_PHY_TS_OA_MY_013_052
ARMOR	Copernicus Marine product: MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_CLIMATE_OHC_IBI_area_averaged_anomalies: it provides annual mean time series of Ocean Heat Content (OHC) averaged over the IBI domain and integrated from surface down to 2000 meters depth.

OMI_CLIMATE_OHC_IBI_area_averaged_anomalies contains 1D (time dimension) OHC information computed from different reanalysis and observational datasets. The product is available for the IBI region (26°N, 56°N; 19°W, 5°E). It is calculated between 0 and 2000 meter depth (see QUID document for more information on the calculation). The interval between each data is one year.

This OMI is computed using IBI-MYP, GLO-MYP reanalysis IBI-INT model dataset, and CORA, ARMOR data from observations which provide temperatures. Where the CMEMS product for each acronym is:

Reference ID	Product information
IBI-MYP (Reanalysis) IBI-INT (Interim dataset)	**CMEMS product:** IBI_MULTIYEAR_PHY_005_002 • PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-IBI-PUM-005-002.pdf • QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-005-002.pdf
GLO-MYP (Reanalysis)	**CMEMS product:** GLOBAL_MULTIYEAR_PHY_001_030 • PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-GLO-PUM-001-030.pdf • QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-GLO-QUID-001-030.pdf
CORA (Observations)	**CMEMS product:** INSITU_GLO_PHY_TS_OA_MY_013_052 • PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-INS-PUM-013-002-052.pdf • QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-INS-QUID-013-052.pdf
ARMOR (Reprocessed observations)	**CMEMS product:** MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 • PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-MOB-PUM-015-012.pdf • QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-MOB-QUID-015-012.pdf

The product is published on the CMEMS dissemination server after automatic and

delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.4 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Product OMI_CLIMATE_OHC_IBI_area_averaged_anomalies is described in the Quality Information Document (QUID).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CLIMATE_OHC_IBI_area_averaged_anomalies
Geographical coverage	26°N, 56°N; 19°W, 5°E
Variables	integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content
Available time series	From 1993 to 31/12/2024
Temporal resolution	Yearly means
Horizontal resolution	N. A. (Area Averaged from diverse resolutions)
Integration depth (for example)	0-2000m
Format	NetCDF4 CF1.7

b) Production System Description

Data from the IBI-MYP and GLO-MYP reanalysis, the IBI-INT model, and the CORA and ARMOR observational datasets are used to derive temperature fields, which serve as the basis for computing the Ocean Heat Content (OHC) anomaly time series. The annual mean time series begins in 1993 and is regularly updated, with each yearly mean centered on June 30. The product provides annual mean files of OHC time series for the Iberia–Biscay–Ireland (IBI) region (26°N–56°N, 19°W–5°E). The OHC values are integrated over the upper 2000 m of the water column.

c) Dataset details

OMI_CLIMATE_OHC_IBI_area_averaged_anomalies contains the 1D annual IBI OHC anomaly from: IBI-MYP, GLO-MYP reanalysis and CORA, ARMOR observations and the ensemble mean as well as the ensemble standard deviation.
ohc_mean [J·m-2] long_name: Ensemble Mean of Area Averaged Ocean Heat Content Anomaly standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content
ohc_std [J·m-2] long_name: Ensemble Standard Deviation of Area Averaged Ocean Heat Content Anomaly standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content
ohc_ibi_myp [J·m-2] long_name: Area Averaged Ocean Heat Content Anomaly from IBI-MYP Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content
ohc_ibi_int [J·m-2] long_name: Area Averaged Ocean Heat Content Anomaly from IBI-MYP Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content
ohc_glo_myp [J·m-2] long_name: Area Averaged Ocean Heat Content Anomaly from GLO-MYP Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content

ohc_cora [J·m-2] long_name: Area Averaged Ocean Heat Content Anomaly from CORA Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content
ohc_armor [J·m-2] long_name: Area Averaged Ocean Heat Content Anomaly from ARMOR Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) REFERENCES

See the Quality Information Document (QUID):
https://documentation.marine.copernicus.eu/QUID/CMEMS-IBI-QUID-CLIMATE-OHC.pdf

von Schuckmann K., Balmaseda M., Simoncelli S. (2016). Ocean Heat Content. In von Schuckmann et al. (2016) The Copernicus Marine Environment Monitoring Service Ocean State Report, Journal of Operational Oceanography, 9:sup2, s235-s320, DOI: 10.1080/1755876X.2016.1273446

von Schuckmann K., Storto A., Simoncelli S., Raj R.P., Samuelsen A., de Pascual Collar A., García Sotillo M. (2018). Ocean Heat Content. In von Schuckmann et al. (2018) The Copernicus Marine Environment Monitoring Service Ocean State Report, Journal of Operational Oceanography, 11:sup1, S1-S142, DOI: 10.1080/1755876X.2018.1489208